SeaIceNet: Sea Ice Recognition via Global–Local Transformer in Optical Remote Sensing Images

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***Abstract*— The recognition of sea ice is of great significance for reflecting climate change and ensuring the safety of ship navigation. Recently, many deep-learning-based methods have been proposed and applied to segment and recognize sea ice regions. However, there are huge differences in sea ice size and irregular edge profiles, which bring challenges to the existing sea ice recognition. In this article, a global–local Transformer network, called SeaIceNet, is proposed for sea ice recognition in optical remote sensing images. In SeaIceNet, a dual global-attention head (DGAH) is proposed to capture global information. On this basis, a global–local feature fusion (GLFF) mechanism is designed to fuse global structural correlation features and local spatial detail features. Furthermore, a detail-guided decoder is developed to retain more high-resolution detail information during feature reconstruction for improving the performance of sea ice recognition. Extensive experiments on several sea ice datasets demonstrated that the proposed SeaIceNet has better performance than the existing methods in multiple evaluation indicators. Moreover, it excels in addressing challenges associated with sea ice recognition in optical remote sensing images, including the difficulty in accurately identifying irregular frozen ponds in complex environments, the broken and unclear boundaries between sea and thin ice that hinder precise segmentation, and the loss of high-resolution spatial details during model learning that complicates refinement.**

***Index Terms*— Deep learning, sea ice recognition, semantic segmentation, Transformer model.**

I. Introduction

SEA ice recognition is critical to understanding the Earth's climate system, predicting weather patterns, protecting ecosystems, ensuring safe navigation, and making informed decisions about resource management and conservation in the face of climate change [1], [2], [3], [4].

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In order to fully study the identification of annual sea ice in coastal zones, it is necessary to select appropriate research data and methods for analysis. Traditional sea ice monitoring methods, such as marine surveillance ships, surveillance aircraft, field surveys, and so on, are difficult to obtain large-area sea ice data due to the limitation of time and space, let alone real-time data update and dynamic monitoring [5]. With the development of remote sensing technology, more and more synthetic aperture radar (SAR) and optical remote sensing data images can be used for sea ice detection and prediction [6]. SAR has strong penetrating ability and is less affected by cloudy and rainy weather, which can provide full polarization and multipolarization data for sea ice monitoring in cloudy and rainy areas, so it is the main way of remote sensing monitoring and recognition of sea ice at present [7], [8], [9], [10]. Compared with SAR data, optical remote sensing will be one of the main data sources for future sea ice research. Although it is greatly affected by weather conditions and is easily covered by clouds, its application cost is low, data acquisition is easy, and the images have higher spatial resolution and are easier for people to understand [11].

In the past research on sea ice recognition by optical remote sensing, exponential segmentation was the main research method [12]. In recent years, deep learning has achieved remarkable success in image semantic segmentation in different fields [13]. Convolutional neural networks (CNNs) can automatically learn features from images and extract high-level semantic information to achieve an accurate understanding of scenes. As a popular image segmentation framework based on CNN, U-Net [14] has the ability to capture image features and perform pixel-level segmentation. Subsequently, Stokholm et al. [15], Huang et al. [16], Wang and Li [17] used U-Net to achieve high classification accuracy in the sea ice SAR image classification task. However, it is difficult to segment and identify optical remote sensing data accurately and efficiently because of the wide distribution of sea ice, the variety of sea ice, and the great influence of background.

On the one hand, there are many factors contributing to the difficulty of sea ice segmentation. In coastal regions, numerous frozen pond formations with diverse shapes are present. These formations are easily affected by various factors, such as local occlusion, lighting, and shadows, which lead to low identification accuracy, as shown in Fig. 1(a). Optical images used for sea ice identification are particularly

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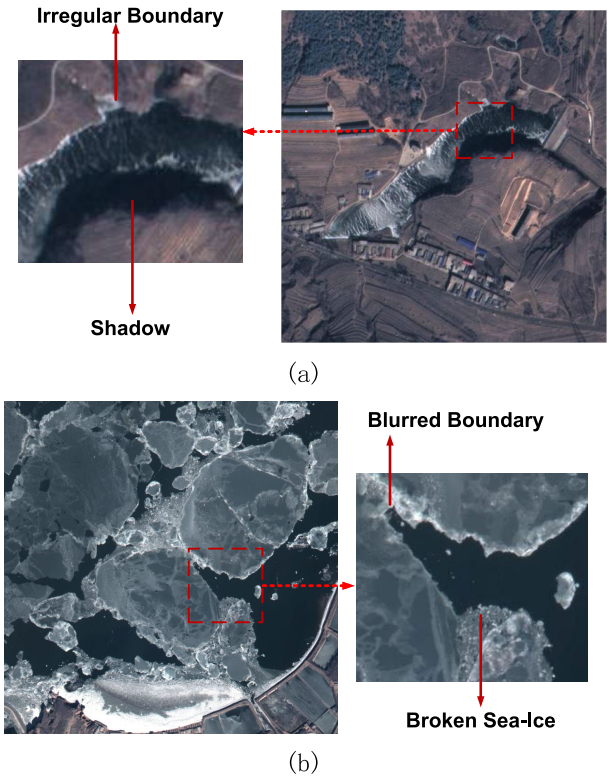


Fig. 1. Challenges in sea ice recognition: (a) irregular frozen pond obscured by shadows and (b) broken sea ice with blurred boundaries.

susceptible to interference from these factors. However, the existing studies, such as the one by Panchi et al. [18], do not adequately address these issues. Their third-order method for segmenting sea ice, including frozen pond, relies solely on close-range local information and cannot be applied to remote sensing images. Therefore, to achieve accurate frozen pond recognition in remote sensing images, it is necessary to perform multidirectional global modeling on the images to address the challenges posed by lighting and morphological variations.

On the other hand, it is difficult to distinguish sea and ice, and thin ice and thick ice when finely dividing sea ice, as shown in Fig. 1(b). Among them, thick ice includes white ice, gray white ice, gray ice, and so on, and thin ice includes ice skin, Nile ice, and so on, as shown in Fig. 2. Thin ice, as the transitional part between thick ice and sea, has a complex form, its boundary is intricate, the shape is changeable, the edge outline is not clear, and the contrast with the surrounding sea or ice is low. It is difficult to accurately identify through a simple visual segmentation method. In contrast, the boundaries of thick ice are often blurred, with gradual or transitional regions, and may have a variety of features, such as ice cubes, cracks, and icebergs, with large-scale differences among these features. Therefore, the fine-grained segmentation of thin ice requires a wealth of detailed information. However, the existing local information extraction module, such as UNetFormer [19], only uses two convolutions to achieve local information extraction, which is mainly designed for the scene of relatively regular ground objects, and cannot well cope with the recognition of thin ice boundaries with broken edges and

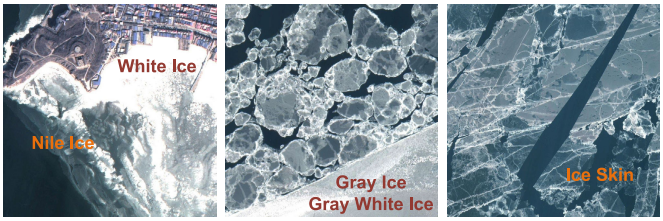


Fig. 2. Schematic of thick ice and thin ice classification.

complexity. Therefore, we urgently need a method that can highlight the edge details of the image to solve the problem of thin ice breaking but not obvious in sea ice segmentation.

In addition, due to the variety of sea ice categories, it is difficult to segment in the identification process, and it is necessary to make full use of the characteristic information. The shallow features generated by ResBlock [20] retain the characteristics of rich spatial details of sea ice, but lack semantic information. In contrast, deep features contain more precise semantic information, but at a lower resolution. At present, the existing encoder–decoder-based segmentation models usually adopt the weighted combination [21], which often ignores important spatial details. Wu et al. [22] and Hong et al. [23] designed a DC-Net architecture and a CCR module, respectively, to incorporate multimodal features, and such modules are typically used for hyperspectral and multispectral images. Therefore, a more effective RGB image fusion strategy is needed to make full use of the advantages of these two features and improve the accuracy of sea ice segmentation and fine-grained recognition.

Moreover, sea ice presents diversity and complexity in remote sensing images, including ice cubes, ice floats, and ice sheets of different densities, thicknesses, and shapes. At present, the SI-STSAR-7 sea ice dataset [24] serves as an example of a dataset primarily composed of SAR satellite remote sensing data. However, its resolution is lower than that of most existing optical satellites due to the limitations of SAR technology. Consequently, these datasets often fail to provide sufficient details. In addition, there are specific challenges related to labeling and identifying broken ice, ice floes, and other types of sea ice. In the field of sea ice recognition, optical remote sensing data have advantages that SAR satellite data cannot compare with, such as high resolution, rich color information, and easy identification and labeling. However, in the existing optical image research on sea ice segmentation, the data used need to be self-labeled during the research process [25]. Therefore, it is urgent to produce an optical remote sensing dataset of sea ice segmentation to carry out identification and provide further data support in this field.

In summary, the existing deep-learning sea ice segmentation methods based on optical satellite data still face the following challenges. On the one hand, there are a large number of breeding ponds near the coastal zone; the frozen pond formed in winter has various forms, and it is difficult to identify the influence factors on land. On the other hand, the sea ice covers a wide range and varies greatly in scale, so it is difficult for the existing methods to accurately segment the sea ice regions of different scales, especially the jagged contour and finely

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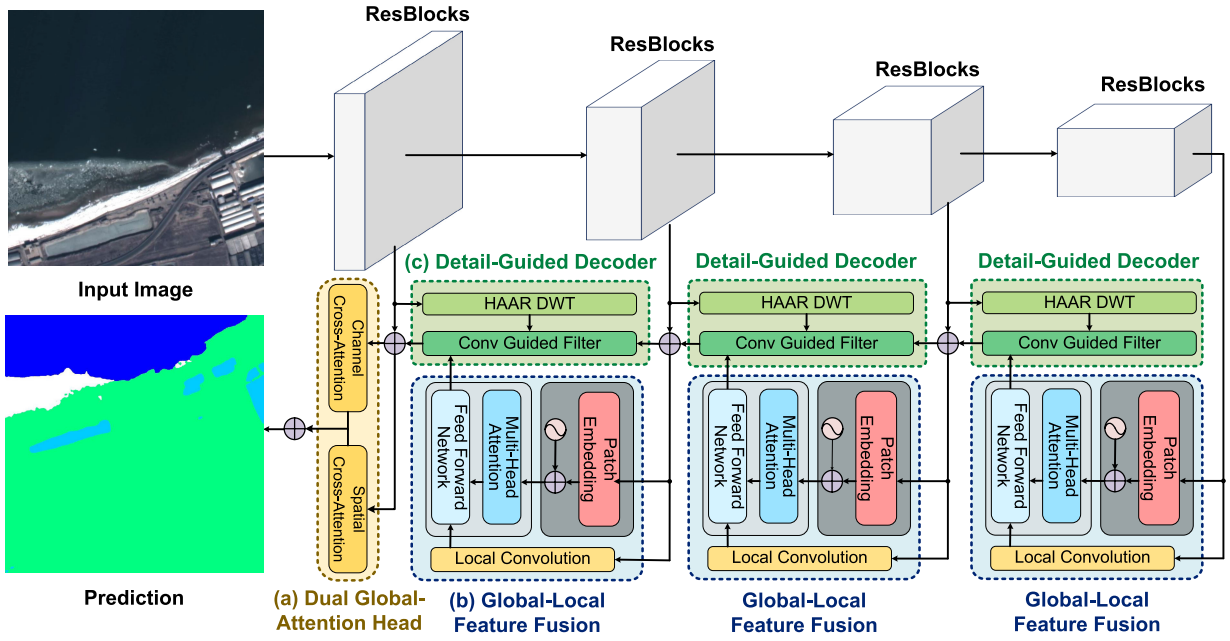


Fig. 3. Framework of the proposed SeaIceNet for sea ice segmentation and recognition. (a) DGAH. (b) DLFF. (c) Detail-guided decoder.

segmented floe ice regions. In addition, the existing methods are not capable of fine-grained identification of different classes of sea ice, especially in areas that are demarcated from different classes of features. Therefore, this article proposes a global–local Transformer network, called SeaIceNet, for sea ice recognition in optical remote sensing images, as shown in Fig. 3. This method can not only reduce the information loss, but also effectively highlight the boundary information. The contributions of this work are summarized in the following.

1) In order to solve the problem that the existing sea ice segmentation methods based on optical data are difficult to obtain global information from multiscale information for frozen pond segmentation, a dual global-attention head (DGAH) is designed. In DGAH, the channel path and spatial path are modeled globally in vertical and horizontal directions to obtain the final global feature representation, so that the onshore frozen pond can be segmented as accurately as possible while avoiding the influence of diverse frozen pond forms.

2) Aiming at solving the problem that the local information extracted by the existing module is not prominent enough, a global–local feature fusion (GLFF) module is proposed in the decoder. The correlation features of the global structure are integrated with the local spatial details to highlight the local high-frequency features while ensuring the integrity of the global information. It can achieve the prominence and recognition of broken thin ice in multiscale sea ice areas.

3) In order to solve the low feature completeness of the existing models, enhance the ability of spatial and semantic information cooperation, and improve the accuracy of model recognition, a detail-guided decoder (DGD) is developed. The learnable filters are guided by the discrete wavelet signature, so that the decoder can take into account the global and local information in the feature reconstruction process, and retain more high-resolution spatial details in the feature reconstruction process. It can improve the recognition performance of fine sea ice.

4) Aiming at addressing the inadequacy of high-resolution optical remote sensing datasets for sea ice segmentation, this study acquired optical remote sensing data from Gaofen series satellites and Google Earth. After the detailed annotation of the images, a fine-grained sea ice segmentation dataset covering five categories was produced. In this work, the validity of the proposed model is verified by this dataset.

The remainder of this article is organized as follows. Section II reviews and analyzes the related works. Section III presents a detailed description of the proposed SeaIceNet. In Section IV, extensive experiments are conducted, and the results are discussed. Finally, conclusions are summarized in Section V.

II. Related Works

A. Cross Attention

In the sea ice segmentation task, the application of cross attention can capture the global correlation information in the input sequence. In general, it is a good solution to use the attention mechanism to capture the deep information of the image, but the use of different attention is also different. The self-attention mechanism is mainly used to capture the long-term dependence between any two positions in the sequence; the multihead attention mechanism is to better capture the association information between different positions in the sequence, and the cross-attention mechanism is used to integrate information from two different sources by introducing additional input sequences to handle some tasks that need to consider the global dependence of two sequences.

Song et al. [26] adopt a combination of axial cross attention and CNN for learning. The axial cross-attention mechanism

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enables the network to effectively capture the association information between different positions in the image. CNN is used for feature extraction and learning. The dual-branch fusion network fuses axial cross attention and CNN features to extract the rich feature information in the image. In the criss-cross attention network (CCNet) designed by Huang et al. [27], the cross-attention module is introduced and constructed. In order to achieve the goal of capturing full-image dependency, simply loop through the cross-concern module. CCNet realizes multilevel integration and utilization of global context information through cross application of attention mechanism in spatial dimension and channel dimension. In addition, Zhu et al. [28] propose pairwise cross attention (PWCA) to establish the interaction between image pairs. PWCA can regularize the attentional learning of an image by using another image as a distracter and is removed during inference. In addition, Ates et al. [29] use dual cross attention (DCA) in their study, which is reflected in the introduction of two cross-attention modules in series in the encoder–decoder structure. DCA addresses the semantic gap between encoder and decoder features by sequentially capturing channel and spatial dependencies between multiscale encoder features.

It can be seen that through cross attention, the model can establish nonlocal correlation dependence, which is particularly effective for dealing with complex scenes, such as frozen ponds with different shapes. Through global modeling, cross attention enables the model to more accurately understand the relationships between the various regions in the sea ice image, thus improving the performance and accuracy of the sea ice segmentation task. However, the last three methods extract the spatial dimension and channel dimension in a series way, and it is easy to lose the information abandoned by the spatial channel when extracting the information of channel dimension. Moreover, the Ates's method extracts the global information of the structure as the skip-connection part of U-Net, which is easily disturbed by the information extracted by the decoder, and cannot fully play the role of cross attention.

B. Gabor Filter

In the broken sea ice scene, local detailed texture information is the key to separating the sea from thin ice and thin ice from thick ice. Gabor filter is a kind of filter with good properties in both the frequency domain and spatial domain, which is usually used to extract the detailed texture and edge features of images. It filters the image in different directions and scales to produce characteristic responses in multiple directions and scales. Therefore, the application of the Gabor filter in sea ice segmentation can obtain the detailed information of multidirection and multiscale sea ice and improve the effect of model learning features.

Chen et al. [30] use the advantage of Gabor filter in frequency analysis; the faster region-based CNN (Faster R-CNN) model is improved by embedding Gabor kernel into Faster R-CNN, and the genetic algorithm is used to determine the optimal Gabor filter parameters. Bhatti et al. [31] proposed local similarity projection Gabor filtering, which aims to simplify a large amount of band information through preprocessing. It reduces the dimension of hyperspectral data by using local similarity analysis and then filters the hyperspectral data by using a 2-D Gabor filter to generate spatial tunnel information. In a follow-up study, Luan et al. [32] propose a method for applying Gabor filters to CNNs, called Gabor convolutional networks (GCNs). The core idea of GCNs is to use Gabor filter in the convolution layer to extract the local texture features of the image, so as to enhance the perception ability of the model to texture, edge, and other details. Among them, Gabor filter has the characteristics of multiscale and multidirection, which can capture texture information of different directions and scales in images. GCNs replaces the traditional convolution kernel by learning the parameters of a set of Gabor filters and uses these Gabor filters in the convolutional layer of the network to carry out convolution operations on the input images. Thus, the model's ability to perceive details, such as texture and edge, is improved.

A large number of scholars have shown that integrating Gabor filters into CNNs can significantly improve the performance of models in texture classification, object detection, and other tasks, especially in processing images with rich textures. Therefore, using the Gabor filter to enhance the texture perception capability of CNNs provides a local detail highlighting method for sea ice segmentation.

C. Guided Filter

Fine detail information is the biggest advantage of high-resolution optical remote sensing images, and preserving the shallow information of images to the maximum extent is of great significance for the fine segmentation of sea ice. The guided filter can adjust the pixel value in the input image according to the information of the image to achieve the effect of smoothing and preserving edges. Through guided filtering, the image details and edge information can be preserved as much as possible while smoothing the image.

He et al. [33] propose a new explicit image filtering guide filter. The guide filter is derived from a locally linear model and generates a filtered output image by considering the content of the guide image. The proposed guide filter can be used as an edge-holding smoothing operator, such as the popular two-sided filter, and has better performance near the edge. In image fusion, Li et al. [34] propose a weighted average technique based on guided filtering to make full use of spatial consistency to fuse the base layer and the detail layer. They first utilize an average filter to obtain a two-scale representation and then fused the base layer and the detail layer by using a weighted average method based on guided filtering. Sun et al. [35] propose a low-light image enhancement algorithm based on improved multiscale retinex and artificial bee colony (ABC) optimization to address the challenges of poor image quality, loss of detail information, and excessive brightness enhancement during image enhancement in low-light environments. The edge-preserving property of the weighted guided image filter is used to enhance the image. In addition, in order to solve the situation where local filtering-based edge retention smoothing techniques can be affected by halo artifacts, Li et al. [36] introduce a weighted

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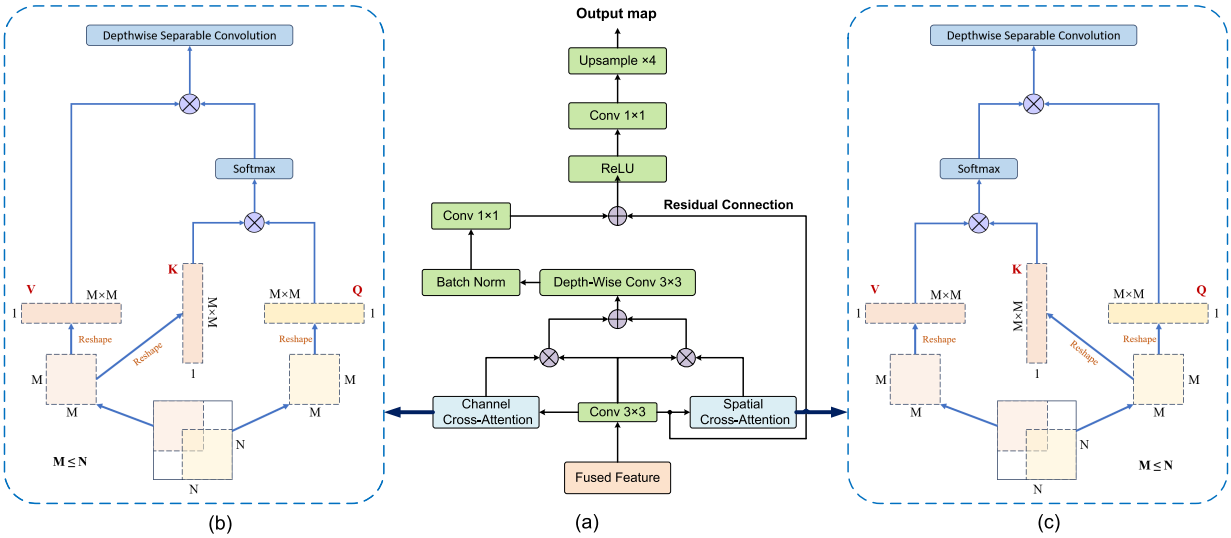


Fig. 4. Architecture of (a) our proposed DGAH block, consisting of (b) channel cross attention and (c) spatial cross attention.

guided image filter to solve this problem by incorporating edge-sensing weights into the existing guided image filters.

In subsequent studies, the guide filter proved to be effective in a variety of computer vision and computer graphics applications, including noise reduction, detail enhancement, high dynamic range compression, image feathering, defogging, and joint upsampling. However, the existing methods are still easy to lose information when applying shallow information and semantic information fusion in guiding images.

III. Proposed GDGT Framework

The proposed SeaIceNet is shown in Fig. 3. The model adopts the U-shaped structure of U-Net, in which the undersampling encoder adopts the ResNet18 structure and the upsampling decoder adopts the Transformer structure. The feature interaction between encoder and decoder is carried out through potential connections.

On the one hand, to enable the segmentation head to obtain enough global vertical and horizontal information, first, DGAH is proposed as the head part of the model to realize the integration of global information. On the other hand, it enables the decoder to learn the global structural features and highlight the local detailed features at the same time and improves the Transformer module of the decoder into, second, a GLFF module to achieve a more comprehensive multiscale feature fusion. In addition, to enable the decoder to retain more high-resolution details during feature reconstruction, third, the DGD module is designed to improve the potential connection between the encoder and the decoder. Discrete wavelet transform (DWT) is used to preserve the high-resolution detailed features of the encoding stage to guide the feature upsampling of the decoding stage. The DGAH, GLFF, and DGD modules are explained in detail below.

A. Dual Global-Attention Head

The cross-attention mechanism extends the self-attention mechanism to capture correlations between different locations in the input sequence more effectively. While self-attention mechanisms primarily focus on relationships within a single input sequence, cross-attention mechanisms enhance this by incorporating additional input sequences. This approach allows for more comprehensive and accurate modeling by fusing information from two distinct sources.

As illustrated in Fig. 4, we designed two channels to enhance feature representation for both channel-type and space-type information. Through the cross-attention fusion mechanism, we effectively learn and integrate these two types of information, achieving a more robust representation. Specifically, we introduce the channel cross-attention (CCA) and spatial cross-attention (SCA) modules. The CCA module extracts channel information, while the SCA module captures spatial information, facilitating global modeling from vertical and horizontal directions, respectively. This design not only enhances the global learning capability and parallelism of the model, but also emphasizes key information in the fusion representation while suppressing irrelevant noise, ultimately improving the model's expressiveness and performance.

In this module, the goal is to process and transform the input data through a series of convolutional and reshaping operations to prepare it for further analysis. First, we perform a 3×3 convolution layer operation on the fused data. This step is crucial, as it helps in capturing spatial relationships and features within the data by applying a set of learnable filters. The convolution operation can be represented as follows:

$$\boldsymbol{F} = \mathrm{Conv2D}_{3\times3}(\boldsymbol{FusedFeature}) \tag{1}$$

where $\boldsymbol{F}$ represents the feature map obtained after applying the 3×3 convolution. After obtaining the feature map from the convolution operation, we proceed to extract patches $\boldsymbol{T_i}$ from the feature map, where i ranges from 1 to n, with n being the total number of patches. This involves dividing the feature map into smaller, nonoverlapping 2-D patches. Each patch is a subregion of the original feature map and is flattened into a 1-D vector. For each extracted and flattened patch $\boldsymbol{E_i}$, we apply a depthwise 1×1 convolution. This projection is to

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transform the flattened patches into a suitable representation for subsequent processing, and it can be described as follows:

$$\boldsymbol{T_i} = \mathrm{DConv1D}_{\boldsymbol{E_i}}(\mathrm{Reshape}(\boldsymbol{E_i})) \tag{2}$$

where $\boldsymbol{E_i}$ is the flattened 2-D patch, $\mathrm{Reshape}(\boldsymbol{E_i})$ converts the patch into a 1-D vector, and DConv1D is the depthwise 1×1 convolution applied to the reshaped patch.

1) Channel Cross Attention: The CCA module, depicted in Fig. 4(b), begins by applying layer normalization to each $\boldsymbol{T_i}$. Following normalization, these patches are concatenated along the channel dimension to form $\boldsymbol{T_c}$, which serves as the basis for generating the key (K) and value (V) components in the self-attention mechanism. Simultaneously, the original patches $\boldsymbol{T_i}$ are used as the query (Q). In this process, the dimensions of Q, K, and V are $\mathbb{R}^{n\times d}$, where d and n represent their respective dimensions and numbers. They can be expressed as follows:

$$\boldsymbol{Q} = \mathrm{DConv1D}_{\boldsymbol{Q}}(\boldsymbol{T_i}) \tag{3}$$

$$\boldsymbol{K} = \mathrm{DConv1D}_{\boldsymbol{K}}(\boldsymbol{T_c}) \tag{4}$$

$$\boldsymbol{V} = \mathrm{DConv1D}_{\boldsymbol{V}}(\boldsymbol{T_c}). \tag{5}$$

The Q, K, and V in the context of the cross-attention mechanism are essential elements that facilitate the calculation of attention scores and the subsequent aggregation of information. Q is used to probe K to determine the relevance or similarity of different pieces of information. The similarity scores, often calculated using dot products, determine how much focus or weight each V should receive. This process can be formally represented as follows:

$$\mathrm{CCA}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{softmax}\left(\frac{\boldsymbol{Q^T K}}{\sqrt{\boldsymbol{d_k}}}\right)\boldsymbol{V^T}. \tag{6}$$

The use of deep separable convolution within the self-attention framework is a strategic choice aimed at capturing essential information while significantly reducing computational complexity. By breaking down the convolution process into a depthwise convolution followed by a pointwise convolution, we maintain the capacity to extract detailed features with fewer computational resources. This attention mechanism yields a weighted sum of the V, where the weights are derived from the similarity between Q and K. The output from this cross-attention process undergoes further processing through depth-separable convolution, ensuring that the final feature representation is both rich in information and computationally efficient. This intricate processing pipeline enhances the model's ability to learn and represent complex patterns effectively.

2) Spatial Cross Attention: The SCA module, illustrated in Fig. 4(c), operates distinctively from the CCA module. In the SCA module, the concatenated $\boldsymbol{T_c}$ is utilized both as the query and key, while each $\boldsymbol{T_i}$ is treated as the value. They can be expressed as follows:

$$\boldsymbol{Q} = \mathrm{DConv1D}_{\boldsymbol{Q}}(\boldsymbol{T_c}) \tag{7}$$

$$\boldsymbol{K} = \mathrm{DConv1D}_{\boldsymbol{K}}(\boldsymbol{T_c}) \tag{8}$$

$$\boldsymbol{V} = \mathrm{DConv1D}_{\boldsymbol{V}}(\boldsymbol{T_i}) \tag{9}$$

$$\mathrm{SCA}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{softmax}\left(\frac{\boldsymbol{Q K^T}}{\sqrt{\boldsymbol{d_k}}}\right)\boldsymbol{V_i} \tag{10}$$

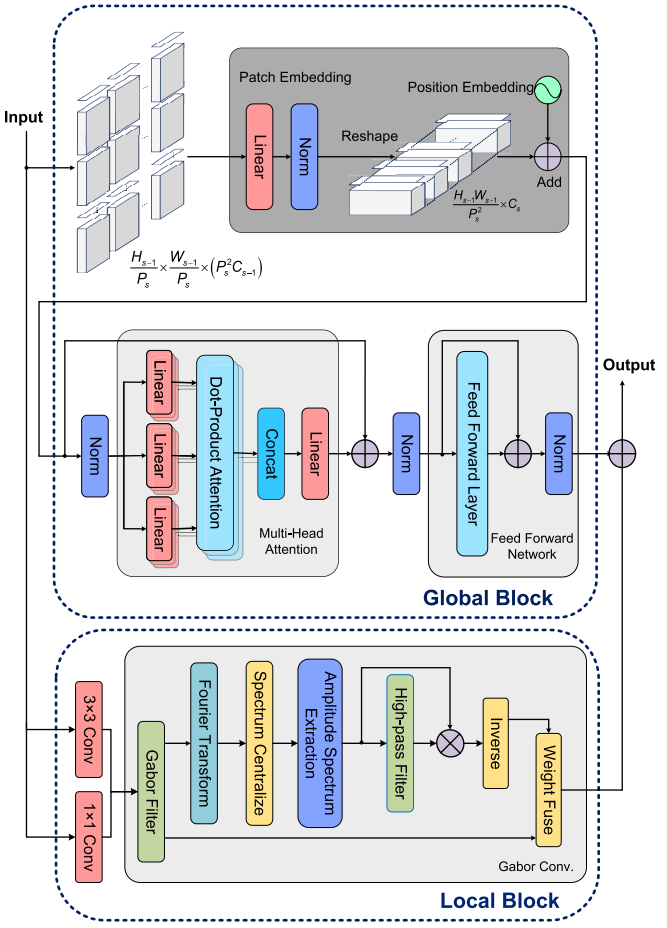


Fig. 5. GLFF module.

Despite this distinction, the subsequent operations remain consistent with those in the CCA module, ensuring a cohesive methodology.

The attention features generated through these parallel pathways are amalgamated using summation operations. This fusion is followed by a 1×1 convolutional layer and an upsampling operation, culminating in the final segmentation predictions. To enhance the model's robustness and mitigate the risk of network degradation, residual connections are incorporated. This design ensures effective feature extraction and maintains high performance throughout the learning process.

B. Global–Local Feature Fusion

Although global context information is very important for semantic segmentation of sea ice scenes, local information has special significance in the field of sea ice segmentation. Local information can enrich the details between the sea ice, so that the model can better distinguish the types of sea ice, such as thick ice, thin ice, and so on.

To solve this problem, we design a new GLFF mechanism, which consists of one global branch, one local branch, two batch normalization layers, and two additional operations, as shown in Fig. 5.

1) Global Branch: Multihead attention is better suited for handling information interactions within the same sequence,

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such as capturing the relationship between semantic information and features inside an image in an encoder. In sea ice segmentation, multihead attention can help the model better understand the texture, edges, and structure information inside the sea ice image, so as to more accurately segment the sea ice area. For this, global branch captures global context with window-based multihead attention. In order to reduce the computational amount of self-attention, the upsampling encoder function is divided into multiple embedded patches. Through the linear layer, the feature vector of each patch is extracted into queries, keys, and values and inputted into the multihead attention block for processing.

2) Local Branch: Local branch from the point of view of image signal, it aims to enhance and extract detailed detail feature information. This branch consists of a Gabor filter and a Fourier transform module, where the Gabor filter adaptively adjusts parameters and simultaneously captures multiple scale features in the image. For sea ice images, texture, structure, and edge features at different scales can be effectively captured by adjusting the scale parameters of the Gabor filter. The Fourier transform, on the other hand, transforms the image from the spatial domain to the frequency domain and can represent the image as a sine and cosine component of different frequencies. This method of spectral analysis can help us understand the contribution of different frequency components in sea ice images and extract important frequency features.

The impulse response of a Gabor filter can be defined as a sine wave multiplied by a Gaussian function. Because of the multiplicative convolution property, the Fourier transform of the impulse response of a Gabor filter is the convolution of the Fourier transform of its harmonic function and the Fourier transform of the Gaussian function. These filters have real and imaginary parts, orthogonal to each other, and arrays of Gabor functions at different frequencies and directions are very efficient for extracting image features. The following is the mathematical expression of the 2-D Gabor function:

Plural expression

$$\boldsymbol{g}(x, y; \lambda, \theta, \psi, \sigma, \gamma) = e^{\left(-\frac{x'^2+\gamma^2 y'^2}{2\sigma^2}\right)} e^{\left(i\left(2\pi\frac{x'}{\lambda}+\psi\right)\right)} \tag{11}$$

where

$$x' = x\cos\theta + y\sin\theta \tag{12}$$

$$y' = -x\sin\theta + y\cos\theta \tag{13}$$

where x and y are the horizontal and vertical coordinate values in the coordinate system; λ is the sine factor wavelength; θ is the angle of the core direction, which specifies the direction of the parallel shift of direction; ψ is the phase offset; γ is the space phase ratio that limits the degree of deviation from circularity; and σ is the standard deviation of Gaussian envelope, which is determined by wavelength and spatial frequency bandwidth. Also, the above parameters are all learnable parameters. Assuming the input image $\boldsymbol{I}(x, y)$, the Gabor-filtered image $\boldsymbol{G_I}(x, y)$ can be expressed as follows:

$$\boldsymbol{G_I}(x, y) = \boldsymbol{I}(x, y) * \boldsymbol{g}(x, y; \lambda, \theta, \psi, \sigma, \gamma) \tag{14}$$

where $*$ indicates a convolution operation.

In the task of sea ice recognition, the use of Fourier transform offers significant advantages over spatial-domain processing for handling detailed features. One of the primary benefits is the ability to effectively separate global and local features. In the spatial domain, processing each pixel directly can be cumbersome, as specific details, such as edges and textures, may get intertwined with the overall brightness or color variations of the image. The Fourier transform addresses this by decomposing the image into different frequency components, thus making it easier to isolate high-frequency components, such as edges and details, from low-frequency components, such as the overall structure. Moreover, frequency-domain processing can enhance contrast and detail.

In this part, given a Gabor-filtered image $G_I(x, y)$ of size $N \times N$, its 2-D discrete Fourier transform $F(u, v)$ is defined as follows:

$$\boldsymbol{F}(u, v) = \sum_{x=0}^{N-1}\sum_{y=0}^{N-1} \boldsymbol{G_I}(x, y) e^{\left(-i2\pi\left(\frac{ux}{N}+\frac{vy}{N}\right)\right)} \tag{15}$$

where u and v are frequency variables and i is imaginary units.

Fourier transform spectrum representation decomposes the spatial-domain signal into a superposition of sinusoidal and cosine waves of different frequencies, revealing the frequency component and intensity distribution of the signal in the frequency domain. Typically, the spectrum obtained by the Fourier transform is symmetric, with the center corresponding to the zero-frequency component, or the direct current (dc) component of the signal. To enhance readability and interpretability, spectrum centralization is performed, moving the zero-frequency component to the center, making the symmetry more apparent and the frequency component distribution clearer.

In sea ice identification, the frequency domain offers significant advantages. It is particularly effective in emphasizing repetitive patterns and textures of different types of sea ice. Thin ice may exhibit smoother textures and more regular frequency components, while thick ice shows complex patterns and a broader frequency range. Due to the complexity and clutter of sea ice images, identifying anomalies, such as fissures, ice ridges, and melt pools, in the spatial domain is challenging. These features, which introduce different frequency components or disrupt regular patterns, are more easily detected in the frequency domain.

After applying the 2-D discrete Fourier transform, the spectrum matrix is divided into four quadrants, each of size $(N/2) \times (N/2)$. Spectrum centralization is achieved by swapping the first quadrant with the third and the second quadrant with the fourth. Mathematically, spectrum centralization can be expressed as follows:

$$\boldsymbol{F_c}(u, v) = \boldsymbol{F}\left(\left(u + \frac{N}{2}\right) \bmod N, \left(v + \frac{N}{2}\right) \bmod N\right) \tag{16}$$

where mod is a modulo operation, which means taking the remainder of the result to ensure that the coordinates are always within the valid range of the image.

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The amplitude spectrum represents the intensity distribution of the frequency components of an image in the frequency domain, providing an intuitive understanding of the frequency-domain features. The amplitude spectrum after Fourier transform can be expressed as follows:

$$\boldsymbol{F_{as}}(u, v) = |\boldsymbol{F_c}(u, v)|. \tag{17}$$

In optical sea ice images, low-frequency information represents the overall structure and large-scale features, while high-frequency information corresponds to details and textural features, such as ice cracks and small fissures. Different types of sea ice exhibit distinct frequency characteristics. Emphasizing high-frequency details and suppressing low-frequency background noise enhance the visualization and analysis of sea ice. To further highlight the high-frequency parts of the image, a high-pass filter can be applied to the amplitude spectrum. The high-pass filter $\boldsymbol{H}$ attenuates or filters out low-frequency components, highlighting high-frequency details and edge information. This can be expressed as follows:

$$\boldsymbol{H}(u, v) = \begin{cases} 0, & \sqrt{\left(u - \frac{N}{2}\right)^2 + \left(v - \frac{N}{2}\right)^2} \le D_0 \\ 1, & \text{otherwise} \end{cases} \tag{18}$$

where D_0 is the cutoff frequency and (u_c, v_c) is the frequency-domain coordinates. After the high-pass filtering operation, the retained high-frequency information will make the details of the image more obvious, thus improving the detection ability of fine structures. It can be expressed as:

$$\boldsymbol{F}_{\text{out}} = \boldsymbol{F_{as}}(u, v) \cdot \boldsymbol{H}(u, v). \tag{19}$$

Next, we will perform inverse spectral centralization and inverse Fourier transform on the spectrum processed by high-pass filtering, respectively, to restore the output results of the local branches. Subsequently, we fused the Gabor results with the frequency-domain augmentation, which could be learned autonomously. It can be expressed as follows:

$$\boldsymbol{out} = \boldsymbol{F}_{\text{out}} \cdot \text{weight} + \boldsymbol{G_I}(x, y) \cdot (1 - \text{weight}). \tag{20}$$

Finally, we can extract local detailed feature information by carrying out maximum pooling and convolution operations to further strengthen the high-frequency details and boundary features of the image. Such manipulation helps to highlight local structure and texture information in the image, making the image sharper and more prominent in terms of detail.

C. Detail-Guided Decoder

The shallow features generated by ResBlock retain rich spatial details of sea ice, but lack semantic information. Deep global–local information has accurate semantic information, but the resolution is low. The common methods are summing or simply weighting the two features directly. Although these methods are fast, they are easy to miss the edge details of the object in the process of downsampling encoding and upsampling decoding of the model. However, fine segmentation of sea ice regions relies on these detailed texture and contour features. Therefore, the DGD module, as shown in Fig. 6, is designed.

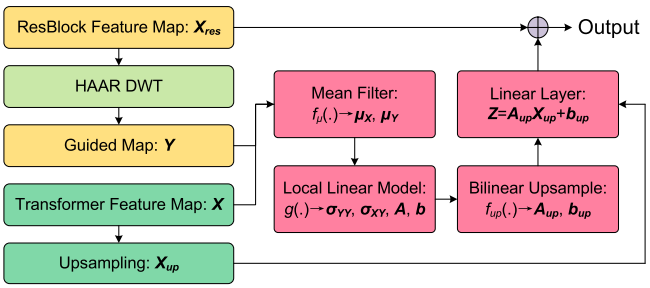


Fig. 6. DGD module.

First, the shallow semantic features with higher resolution are decomposed by HAAR wavelet to obtain high-frequency detailed features. Denote the shallow semantic features from a residual block of the encoder as $\boldsymbol{X_{res}}$, which have

$$\boldsymbol{LL}, \boldsymbol{LH}, \boldsymbol{HL}, \boldsymbol{HH} = dwt(\boldsymbol{X_{res}}) \tag{21}$$

where $dwt(\cdot)$ represents the discrete HAAR wavelet transform. The low-frequency component $\boldsymbol{LL}$, horizontal high-frequency component $\boldsymbol{LH}$, vertical high-frequency component $\boldsymbol{HL}$, and diagonal high-frequency component $\boldsymbol{HH}$ are obtained by HAAR wavelet decomposition

$$\begin{cases} LL_{i,j} = X_{2i-1,2j-1} + X_{2i-1,2j} + X_{2i,2j-1} + X_{2i,2j} \\ LH_{i,j} = -X_{2i-1,2j-1} - X_{2i-1,2j} + X_{2i,2j-1} + X_{2i,2j} \\ HL_{i,j} = -X_{2i-1,2j-1} + X_{2i-1,2j} - X_{2i,2j-1} + X_{2i,2j} \\ HH_{i,j} = X_{2i-1,2j-1} - X_{2i-1,2j} - X_{2i,2j-1} + X_{2i,2j} \end{cases} \tag{22}$$

where $LL_{i,j}$, $LH_{i,j}$, $HL_{i,j}$, and $HH_{i,j}$ represent the elements of $\boldsymbol{LL}$, $\boldsymbol{LH}$, $\boldsymbol{HL}$, and $\boldsymbol{HH}$ at (i, j), respectively.

Then, through the learnable guided filtering mechanism, detailed features are retained and enhanced in the decoder, improving the model's segmentation accuracy for sea ice areas (especially, fine sea ice areas and thin ice areas). Denoting the guided feature map containing rich detailed features of sea ice images as

$$\boldsymbol{Y} = \text{Conv}(\boldsymbol{LL}, \boldsymbol{LH}, \boldsymbol{HL}, \boldsymbol{HH}) \tag{23}$$

where $\text{Conv}(\cdot)$ denotes the 1×1 convolution, is generated

$$\boldsymbol{\mu_X} = f_\mu(\boldsymbol{X}), \quad \boldsymbol{\mu_Y} = f_\mu(\boldsymbol{Y}) \tag{24}$$

where f_μ denotes a learnable mean filter constructed through convolutional layers

$$\boldsymbol{\sigma_{XY}} = g(\boldsymbol{X} \cdot \boldsymbol{Y}), \quad \boldsymbol{\sigma_Y} = g(\boldsymbol{Y} \cdot \boldsymbol{Y}) \tag{25}$$

where g denotes a local linear operation

$$g(\mathbf{X} \cdot \mathbf{Y}) = f_\mu(\mathbf{X} \cdot \mathbf{Y}) - \boldsymbol{\mu_X} \cdot \boldsymbol{\mu_Y} \tag{26}$$

$$\boldsymbol{A} = \frac{\boldsymbol{\sigma_{XY}}}{\boldsymbol{\sigma_Y} + \varepsilon}, \boldsymbol{b} = \boldsymbol{\mu_Y} - \boldsymbol{A} \cdot \boldsymbol{\mu_X} \tag{27}$$

where ε denotes the learnable regularization term. Thus, the output of the DGD module is

$$\boldsymbol{Z} = \alpha \cdot \text{up}(\boldsymbol{A}) \cdot \text{up}(\boldsymbol{X}) + \text{up}(\boldsymbol{b}) + \beta \cdot \boldsymbol{X_{res}} \tag{28}$$

where $\text{up}(\cdot)$ represents the upsampling operation, and α and β denote the learnable weights.

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IV. Experiments and Discussions

A. Experimental Conditions

1) Experimental Details: All experiments were performed on a computer with an RTX4070 GPU or an RTX4090 GPU. For fast convergence, we deployed the AdamW optimizer to train all the models in the experiment. The maximum training period is set to 100, and the initial learning rate is set to 6×10^{-4}. This learning rate was adopted in the original model and the original control model, and we did not make any special adjustments to ensure the fairness of the training process. Different networks do differ in the convergence speed. We initially used 300 epochs, and after observing the change curve of mIoU, we selected the fastest convergence range and finally settled on 100 epochs.

2) Evaluation Index: The evaluation metrics used in our experiments included two major categories. The first one was to evaluate the accuracy of the network, including the overall accuracy (OA), mean $F1$ score ($F1$), and mean intersection over union (mIoU). In addition, in order to eliminate the problem of uneven distribution of sample pixels in multiclassification problems, we also use fully weighed intersection over union (FWIoU) to evaluate the model. The second one was to evaluate the scale of the network, including the memory footprint (MB) and the number of model parameters (M) to evaluate the memory requirement.

3) Baseline: Baseline is UNetFormer [19] proposed by Wang et al., which uses ResNet18 as the backbone of the model and Transformer as part of the encoder. To improve model performance, we have tried other ResNet backbone, such as ResNet32, ResNet50, and ResNet101. At the same time, we also tried a number of backbones, including ResNet and large selected kernel, but compared with a large increase in the size of the model, the performance improvement is not obvious. Therefore, considering the efficiency and performance of the model, we finally chose ResNet18, which is adopted by baseline, as the backbone of our model.

4) Pretrained: In our initial experiments with the GFGE dataset, we also validated the effects of using a pretrained model versus an untrained model. We found that for this dataset, the pretrained model only affected the learning speed of the model, but did not affect the best performance of the model. Therefore, we used the pretrained model for all models, including the control model in the subsequent experiment.

B. Experiments Based on the GFGE Dataset

We collected GF series optical remote sensing satellite and Google Earth image data and completed the sea ice segmentation dataset on this basis. The original image of this dataset was taken on January 28, 2022, in the eastern part of Liaodong Bay in China's Bohai Sea, and all sea ice is annual sea ice. To enable the model to capture multiscale features of the image at different levels of detail, it scales at different resolutions: $\times 0.25$, $\times 0.50$, $\times 1.00$, and $\times 1.50$. Due to the large size of the remote sensing images, they are cropped into subimages of 1024×1024 pixels (overlapping 256 pixels).

The images are carefully annotated at the pixel level. There are five categories of objects that need to be identified, including "sea," "thin ice," "thick ice," "land," and "frozen pond." In this dataset, "sea," "thin ice," "thick ice," "land," and "frozen pond" account for 32.5%, 10.7%, 17.3%, 36.2%, and 3.3%, respectively. When separating the training set and the verification set, we ensure that the proportion of each element is as close as possible to the total proportion. When the deep-learning model was input for training and testing, the training and testing datasets contained 8178 and 1980 pairs of images and truth values, respectively.

TABLE I
Ablation Study of Each Component of the SeaIceNet

Methods	DGD	GLFF	DGAH	FWIoU(%)	mIoU(%)	OA(%)	F1(%)
UNetFormer[19]				83.4663	74.9681	89.9636	82.9548
w. DGD	✓			83.9742	75.8705	90.2550	83.8927
w. GLFF		✓		84.4711	75.7429	90.5228	83.7497
w. DGAH			✓	84.2823	75.8695	90.0698	83.8842
w/o. DGAH	✓	✓		84.8549	76.5361	91.0817	84.6725
w/o. GLFF	✓		✓	83.0433	76.3110	89.6959	84.7425
w/o. DGD		✓	✓	84.2207	76.7591	90.5189	84.8647
SeaIceNet	✓	✓	✓	**86.2797**	**77.1320**	**91.8405**	**84.9063**

1) Ablation Study: To evaluate the performance of each module of the proposed SeaIceNet separately, we performed a series of ablation experiments on the GFGE dataset. The experimental results are shown in Table I. Each of our modules can increase the mIoU of the model in the dataset by 0.7748%. The complete SeaIceNet is 2.1639%, 1.8769%, 1.9515%, and 2.8134% higher than the mIoU, OA, $F1$ score, and FWIoU of baseline UNetFormer, respectively, showing good SeaIceNet performance.

2) For the DGAH Module: It integrates spatial and channel information through intersection, which effectively improves the model's ability to capture complex structures and details. Specifically, the dual global-attention first performs cross-attention calculations in the spatial and channel dimensions of the input features and then fuses the results of the two to comprehensively consider the spatial and channel information at each position of the feature map. This mechanism is particularly well suited for working with images with complex structures and multiscale features, such as land pond ice in sea ice.

In contrast, SE Attention [37] weights each channel globally to improve the feature expression of important channels. Although SE Attention accurately models the relative importance of channels, it cannot directly capture spatial information and has limited ability to enhance spatial details. Multihead attention [38] is derived from the Transformer structure, which captures a variety of different representations of feature maps through the self-attention mechanism of multiple heads. Although multihead attention performs well in natural language processing and image processing tasks, it has a high computational complexity and is not specially designed for the fusion of spatial and channel attention. The spatial attention mechanism [39] highlights the important spatial positions by weighting the spatial dimension of the feature map, which enhances the model's perception of spatial layout,

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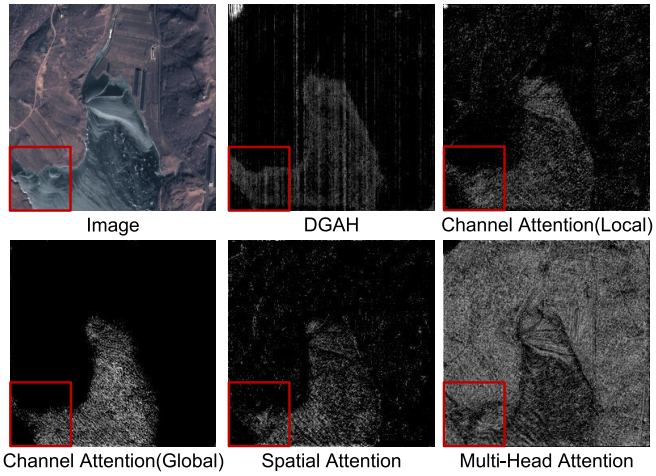


Fig. 7. Comparison of renderings with different attention spans in DGAH.

but ignores the information interaction between channels and cannot capture the global channel relationship.

The channel attention mechanism adopted comes from CBAM [40], and its idea comes from SE Attention, but there are differences between them. SE Attention is local channel attention, while CBAM channel attention is global channel attention. To make this distinction easier, we call them channel attention (local) and channel attention (global). In this part, we perform a substitution comparison of cross attention in DGAH, and the results are shown in Fig. 7.

We can see that DGAH has less noise and clearer boundaries than channel attention (local). Although channel attention (global) has a good noise cancellation effect in the land part, the representation effect of the frame position in the figure is insufficient, and it is difficult to accurately segment it. However, spatial attention and multihead attention are noisy compared with the other attention mentioned earlier, and it is difficult to accurately divide the ice part of the pond. So, the effect of using dual global attention here is optimal.

3) For the GLFF Module: We apply a high-pass filter to the frequency-domain information in the local branch to further enhance the details of the image. A high-pass filter emphasizes the edges and details of an image by retaining high-frequency components and suppressing low-frequency components. In our experiment, we analyzed the effect of different cutoff frequencies every 10 from 0 to 100 to analyze the effect on the image effect.

Through a series of experiments, we compared the image results at different cutoff frequencies, as shown in Fig. 8. Different cutoff frequencies have a significant impact on the detail enhancement effect and noise suppression ability of the image. When the cutoff frequency is low, the high-pass filter mainly retains the detail information of the lower frequency, and the image noise is less, but the edge features are not prominent enough to distinguish the sea from the thin ice. When the cutoff frequency is high, the high-pass filter retains more high-frequency detail information, and the edge and detail features of the image are more obvious, but the noise also increases, and it is difficult to distinguish thin ice from thick ice.

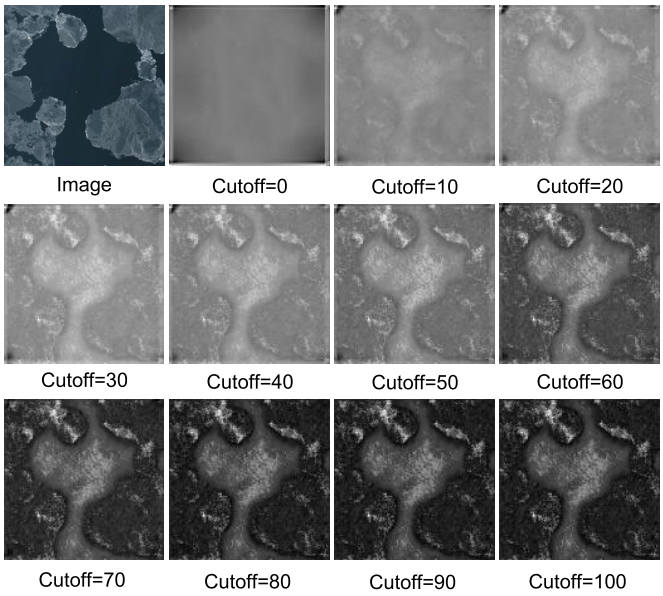


Fig. 8. Visualization of the GLFF module's high-pass filter cutoff frequency sensitivity test.

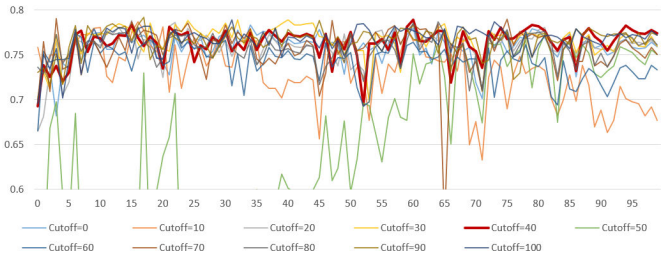


Fig. 9. High-pass filter cutoff frequency sensitivity test of the GLFF module.

As shown in Fig. 9, we train the model with different cutoff frequencies to obtain a series of mIoU trends. We can find that the OA is poor when the cutoff frequency is low or high, and although some of the training results reach a normal level at the end under the effect of weight fuse, the stability is poor. Therefore, in this experiment, we set the cutoff frequency to 40. We did not update this hyperparameter on other datasets in the future.

At the end of the GLFF module, we extract local detail feature information through convolution operation to further enhance the high-frequency detail and boundary feature of the image. Here, the convolution kernel adaptively extracts important information from the image by training the sample. As shown in Fig. 10, it can be found that after such learnable convolution operation, the separation of sea ice boundaries becomes more obvious.

In the specific sea ice segmentation task, the processed images are input into the segmentation network for training, and the segmentation results show higher accuracy in the details and edges, especially in complex backgrounds and areas with rich details. As shown in Fig. 11, the three markers in the figure are the parts where the sea is separated from the sea ice, and by using GLFF, we can see the boundary between the sea and the sea ice more clearly, which is conducive to fine segmentation.

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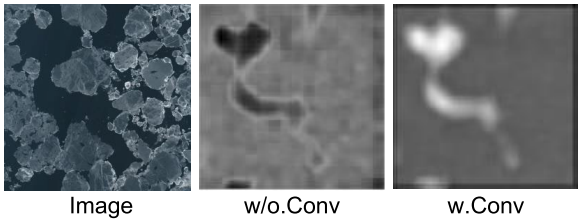


Fig. 10. Convolution operation effect diagram.



Fig. 11. GLFF module visualization.

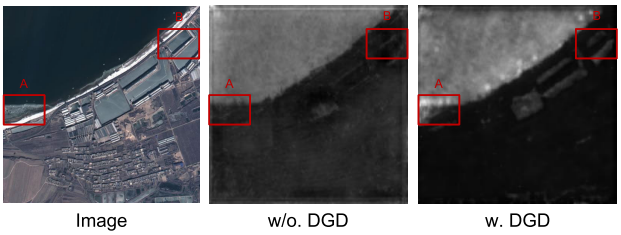


Fig. 12. DGD module visualization.

4) For the DGD Module: It can effectively enhance the information retention of the image, thereby improving the accuracy of segmentation. We have added a DGD module to the front end of the network decoder, so that the network can obtain richer and clearer feature information in the decoding stage. In the process of training and testing the network, this combination method significantly improves the performance of segmentation, especially in the part of information retention. In order to prove the validity of this module, we compare the effect of this module on model interpretation through experiments, as shown in Fig. 12. In area A, the model without DGD can only separate the sea, while the model with DGD can make a clear distinction among sea, sea ice, and land. In area B of the figure, the model using this module is able to retain the pond information well. The DGD module can improve the accuracy and robustness of segmentation while maintaining image information.

When analyzing the results of the ablation experiments of each model in Fig. 13, it can be seen that the DGAH, GLFF, and DGD modules play a crucial role in the model performance.

The DGAH module is mainly used to capture global information to solve the problem that the ice edge of the onshore pond cannot be accurately identified due to the influence of irregularity, shading, and so on. By comparing with the model without the DGAH module, the results show that in the model with the DGAH module, the identification of the pond ice edge is more accurate, the edge contour is clearer, and the area affected by the shadow is also better processed, and the overall recognition effect is significantly improved.

The GLFF module is designed to solve the problem that it is difficult to accurately identify the detailed feature information of sea ice. By comparing the model output with or without the GLFF module, it can be observed that the model with the GLFF module has a more detailed performance in detail features, and the edge of the sea ice has been more accurately restored, especially in the distinction between thin ice and sea, and the details are more in place, which shows the effectiveness of the GLFF module in enhancing the recognition of detail features.

The DGD module is used to solve the problem of information loss in deep-learning models. In the absence of the DGD module, the model performs poorly in complex environments and is prone to recognition errors caused by information loss. However, after adding the DGD module, the model can better retain and convey key information, making the identification results more accurate and stable, especially in the pond ice distinction between different types of ice surface and land areas, showing the advantages of the DGD module in information retention and enhancement.

Through the visual comparison of these ablation experiments, the advantages of the SeaIceNet model are fully reflected. The complete SeaIceNet model has excellent performance in recognition accuracy, detail processing, and information preservation and comprehensively solves the problems of irregular ice edge in land ponds, difficulty in identifying sea ice details and information loss in deep-learning models, and the comprehensive performance is better than that of removing any one module, which proves the importance and effectiveness of each module in the overall model.

5) Result on the GFGE Dataset: The GFGE dataset is the sea ice remote sensing dataset we produce, which is divided into five categories, and it is difficult to achieve overall high-precision classification in optical remote sensing sea ice classification and multiclassification tasks. To compare the effectiveness of our models, we trained several advanced efficient segmentation networks and compared the results in detail in the dataset. As shown in Table II, our proposed method yields the best mIoU, FWIoU, OA, and $F1$ scores. Specifically, the proposed SeaIceNet is not only 1.4% higher than the CNN-based efficient network ABCNet on mIoU, but also 1.3% and 2.2% higher than the Transformer-based hybrid networks BANet and UNetFormer, respectively.

In addition, the visualization of the GFGE dataset, as shown in Fig. 14, also demonstrates the effectiveness of our SeaIceNet. In the first set of images, although U-Net can identify most of the sea ice areas, there is a lack of key feature information in the sea ice detail processing, which is easy to confuse some pixels in the thick ice with thin ice. Compared with U-Net, ABCNet has improved the information processing within the global network, but there are still some problems of missegmentation of details, especially in the complex background of thick ice and thin ice. BANet has improved the segmentation accuracy to a certain extent, but there are still deficiencies in the image processing of the displayed

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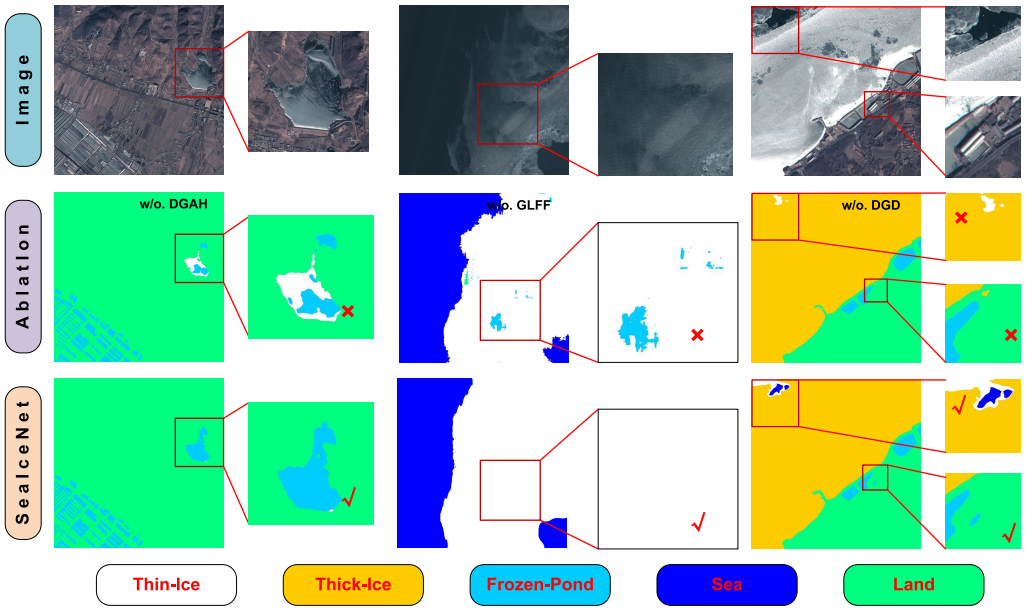


Fig. 13. Ablation study results from the GFGE dataset.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT SEMANTIC SEGMENTATION METHODS ON THE GFGE DATASET

Method	Backbone	Sea(%)	Thin Ice(%)	Thick Ice(%)	Frozen Pond(%)	Land(%)	FWIoU(%)	mIoU(%)	OA(%)	F1(%)	Para.(M)	Size(MB)
U-Net[14]	VGG16	86.7334	67.6034	**41.9793**	57.5237	98.0387	85.0883	75.2351	91.5043	84.0851	31.0	124.15
ABCNet[41]	ResNet18	85.1244	57.4570	38.8869	74.6877	98.7457	82.6009	75.7378	89.4513	84.2641	14.0	55.85
BANet[42]	ResT-Lite	86.8103	66.6273	26.7308	76.4030	98.7952	84.7396	75.8162	90.7734	83.4797	12.7	50.92
DC-Swin[43]	Swin-Small	86.2440	67.4705	35.7699	46.6787	98.2639	84.6142	72.3242	90.9307	81.4021	66.9	267.80
MANet[44]	ResNet50	88.1194	59.4911	35.2110	68.6005	98.6922	83.8219	74.9414	89.7321	83.4756	35.9	143.44
UNetFormer[19]	ResNet18	83.8673	65.6811	26.7165	75.4572	98.5576	83.4663	74.9681	89.9636	82.9548	11.7	46.91
FT-UNetFormer[19]	Swin-Base	78.5505	61.4082	36.2589	**77.6114**	**98.8264**	81.2399	75.3652	88.5038	83.9783	37.5	150.07
SeaIceNet	ResNet18	**88.7707**	**69.8504**	33.1052	72.8418	98.6887	**86.2797**	**77.1320**	**91.8405**	**84.9063**	12.5	50.07

images, and it is easy to miss or misdistinguish sea ice features. In the first set of images, SeaIceNet, through its DGD module, effectively improves the ability to recognize detailed features, with clear boundaries and rich details, especially when dealing with onshore pond ice, which can accurately segment sea ice areas in complex backgrounds.

In the second set of images, DC-Swin performs well in global information capture, but it still lacks in detail processing and is prone to blurred boundaries and misdivision of small ice blocks. MANet has improved in detailing, but there is still a certain degree of missegmentation when separating thin ice and land. UNetFormer performed slightly better than the first two in complex backgrounds, but there were still some inaccuracies in detail and boundary processing, especially in extreme conditions. The overall performance of FT-UNetFormer was better in the second group, but it was easy to mistake snow on land as ice, which was due to the lack of global and local fusion information. In the second set of images, SeaIceNet further improves the ability to capture details by combining multihead attention and Gabor filtering in the frequency domain. It excels in the identification of thin ice, with precise boundary processing, sharp detail, and high segmentation accuracy against complex backgrounds.

U-Net, ABCNet, BANet, and UNetFormer still have the problems of insufficient detail processing, blurred boundaries, and missegmentation in complex backgrounds when processing the third group of images. Although DC-Swin, MANet, FT-UNetFormer can identify the whole range of frozen ponds well, they are prone to leakage when there are shadows and color changes in the ponds. However, the performance of SeaIceNet is still outstanding, and its DGAH module enables the model to achieve a better effect in global information extraction and processing in complex environments, especially in shadow conditions. In conclusion, SeaIceNet outperforms other models in images, especially in the task of sea ice segmentation in detail processing and complex backgrounds, and SeaIceNet significantly improves the segmentation accuracy and stability through its innovative combination of DGAH, GLFF, and DGD.

C. Experiments Based on the HY Dataset

The dataset is a sea ice segmentation dataset [45] based on optical remote sensing images provided by the 2021 Gaofen Challenge, which is composed of Ocean 1 optical satellite data with a resolution of 50 m. Due to the low resolution

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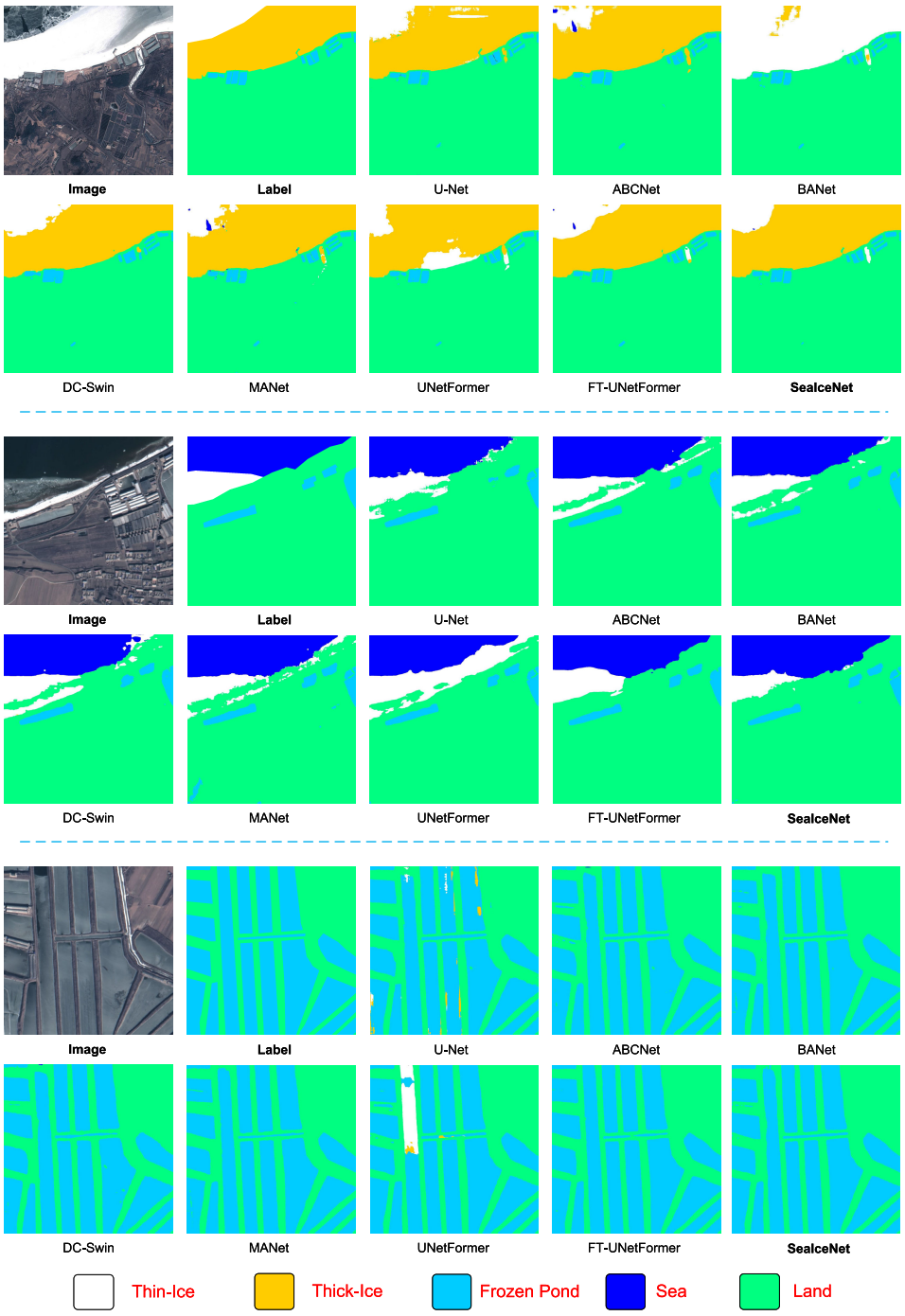


Fig. 14. Controlled trial on the GFGE dataset.

of the image, the labels are only divided into two categories, including "sea ice" and "background." In order to increase the number of datasets, the image is enhanced by flipping, increasing noise, and increasing or decreasing contrast and saturation in advance. When the deep-learning model was input for training and testing, the training and testing datasets contained 6678 and 1794 pairs of images and truth values, respectively.

For a comprehensive evaluation, experiments are also performed on the HY dataset. As shown in Table III, SeaIceNet achieves 99.1639% accuracy in sea ice classification and 90.3889% accuracy in background classification. In addition, SeaIceNet attains key performance metrics of mIoU, FWIoU, OA, and $F1$ at 94.7764%, 98.4978%, 99.2248%, and 97.2660%, respectively, demonstrating superior performance. In comparison, other models fall short in these metrics. For instance, U-Net achieves 98.9114% and 87.5033% accuracy in sea ice and background classifications, with mIoU and $F1$ at 93.2074% and 96.3940%. ABCNet also lags behind SeaIceNet, with sea ice and background classification accuracies at 98.7924% and 86.2486%, and mIoU and $F1$ at 92.5205% and 96.3900%. SeaIceNet's results are not only

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TABLE III
PERFORMANCE COMPARISON OF DIFFERENT SEMANTIC SEGMENTATION METHODS ON THE HY DATASET

Method	Backbone	SeaIce(%)	Background(%)	mIoU(%)	FWIoU(%)	OA(%)	F1(%)
U-Net[14]	VGG16	98.9114	87.5033	93.2074	98.0455	98.9885	96.3940
ABCNet[41]	ResNet18	98.7924	86.2486	92.5205	97.8402	98.8774	96.0046
BANet[42]	ResT-Lite	99.1420	90.1372	94.6396	98.4584	99.2043	97.1910
DC-Swin[43]	Swim-Small	98.4081	83.0963	90.7522	97.2458	98.5236	94.9828
MANet[44]	ResNet50	99.1400	90.0882	94.6141	98.4528	99.2023	97.1769
UNetFormer[19]	ResNet18	99.1096	89.8574	94.4835	98.4073	99.1747	97.1053
FT-UNetFormer[19]	Swin-Base	99.1264	90.0088	94.5677	98.4343	99.1902	97.1515
SeaIceNet	ResNet18	**99.1639**	**90.3889**	**94.7764**	**98.4978**	**99.2248**	**97.2660**

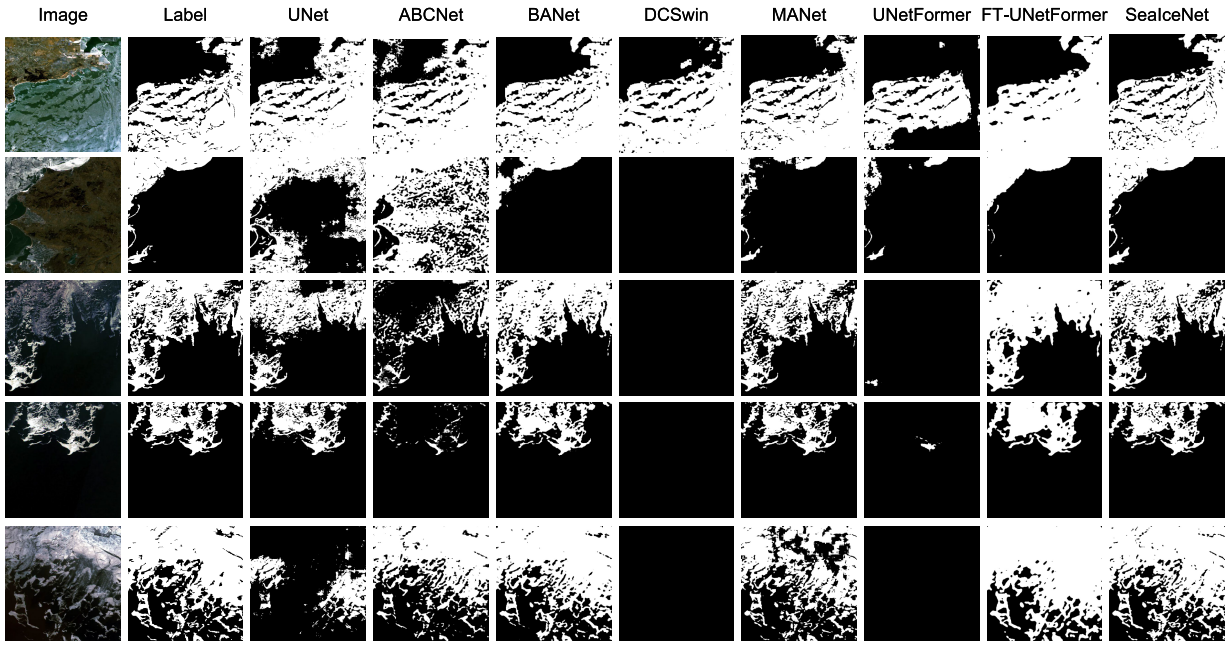


Fig. 15. Controlled trial on the HY dataset.

superior to ABCNet, an excellent lightweight network, but also to Transformer-based DC-Swin. Compared with DC-Swin, SeaIceNet is 0.7558%, 7.2926%, 4.0242%, and 2.2832% higher on sea ice, background, mIoU, and $F1$, respectively.

The visualization is shown in Fig. 15. The U-Net model showed basic ability to segment the sea ice from the background, but there were obvious deficiencies in the details. The segmentation of the edge part is blurred, and it is easy to confuse the sea ice with the background area in the complex background, resulting in the inaccurate identification of some sea ice areas, such as the second- and fifth-group experiments. Compared with U-Net, ABCNet is generally similar, but there are still problems in the recognition of complex edges and shapes. While there are some areas where the sea ice and background are well separated, the overall edge sharpness and detail treatment need to be improved, as shown in the second group and only the fourth group. BANet has shown good ability in segmentation of sea ice, especially in the identification of sea ice margins. However, in some complex background cases, as shown in the second group, BANet still has cases of misidentification, resulting in some sea ice being incorrectly segmented into the background. Although the DC-Swin model can extract features at a deeper level, there is still a large amount of information loss in some areas, which makes it difficult to identify the internal structure of sea ice. MANet does a good job of handling detailed features, being able to identify the fine structure of sea ice. Despite this, the performance of MANet is not stable in the segmentation of the marginal zone, and some parts of the sea in contact with sea ice are still vague and inaccurate.

UNetFormer combines the basic structure of U-Net with the characteristics of Transformer to improve the overall effect of segmentation, but the performance is extremely unstable in some scenarios, and there is a large area of misfraction. FT-UNetFormer has shown better performance in sea ice segmentation tasks, especially in global information capture. Although the overall effect is good, there will still be the problem of insufficient detail processing. SeaIceNet is the best of all models. It outperforms other models in edge sharpness, detail preservation, and global information capture. SeaIceNet is able to accurately segment sea ice from the background, especially when working with complex edges and

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TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT SEMANTIC SEGMENTATION METHODS ON THE SI-STSAR-7 DATASET

Method	Backbone	GI(%)	GWI(%)	MedFI(%)	NI(%)	OW(%)	ThickFI(%)	ThinFI(%)	FWIoU(%)	mIoU(%)	OA(%)	F1(%)
U-Net[14]	VGG16	12.1330	24.3855	38.8085	1.6179	12.3086	3.1044	35.0773	20.6318	18.2050	37.8511	28.5469
ABCNet[41]	ResNet18	**99.9491**	**99.9999**	81.9098	99.8877	99.9431	73.7312	99.8101	92.7522	93.6044	95.9042	96.3900
BANet[42]	ResT-Lite	99.4939	**99.9999**	86.2026	99.4495	99.9976	83.5258	97.6463	94.4560	95.1879	97.0189	97.4131
DC-Swin[43]	Swin-Small	54.9011	62.6955	58.9880	62.6404	82.0536	53.5386	72.8983	63.7287	63.9594	77.3696	77.6281
MANet[44]	ResNet50	26.2656	14.6005	49.7793	51.5266	85.4302	44.6790	53.3419	46.7299	46.5176	62.4001	60.7204
UNetFormer[19]	ResNet18	99.9311	99.9660	90.7430	**99.9442**	**99.9999**	86.9276	**99.9963**	96.3488	96.7869	98.0499	98.2960
FT-UNetFormer[19]	Swin-Base	37.5997	53.3464	58.5502	13.2234	63.1986	7.3521	74.8350	46.3466	44.0151	65.6780	56.8851
SeaIceNet	ResNet18	99.6464	99.8338	**94.4655**	99.5568	99.8859	**91.7393**	99.7037	**97.5884**	**97.8330**	**98.7538**	**98.8797**

fine structures. Because of the combined effect of DGAH, GLFF, and DGD modules, SeaIceNet not only solves the problems of irregular sea ice edges, difficult identification of detailed feature information, and loss of information in the depth learning model, but also significantly improves the OA and stability of segmentation.

D. Experiments Based on the SI-STSAR-7 Dataset

Although our modules are proposed for optical data, they are also suitable for SAR data. The proposed DGD module can retain more properties, such as object structure and roughness reflected by SAR images. The GLFF module enhances the common edge features in SAR images, highlighting the sea ice boundary. In addition, the DGAH module further improves the ability to extract directional features in images by capturing channel and spatial orientation information.

SI-STSAR-7 [24] is a labeled spatiotemporal dataset for sea ice classification based on SAR images. The dataset is produced from 80 Sentinel-1 A/B SAR scenes during the two freezing periods of Hudson Bay from October 2019 to May 2020 and from October 2020 to April 2021, which are provided by the Copernicus Open Access Center. The dataset includes seven classes, including open water (OW), new ice (NI), gray ice (GI), gray white ice (GWI), thin first year ice (ThinFI), medium first year ice (MedFI), and thick first year ice (ThickFI). When the deep-learning model was input for training and testing, the training and testing datasets contained 786 432 and 196 608 pairs of 32 × 32 images and truth values, respectively.

In order to verify that our model has good results for the segmentation of sea ice in different scenarios, we also conducted a series of control experiments on the sea ice dataset SI-STSAR-7 of SAR images, as shown in Table IV. SeaIceNet significantly outperforms other models in terms of overall performance. On the mIoU and $F1$ scores, SeaIceNet reached 97.8330% and 98.8797%, respectively, which was significantly higher than other models. This shows that SeaIceNet has a significant advantage in terms of overall performance and accuracy.

Through the experimental results, we can see that the recognition effect of different models in this dataset is quite different, and the performance effect of ABCNet, BANet, UNetFormer, and our proposed SeaIceNet is much higher than that of other models. From the perspective of different types of recognition, each model has the best overall recognition effect in the recognition of GI, GWI, NI, and OW, which reflects that these four categories belong to the easier to distinguish in the recognition process. Among them, ABCNet, BANet, and UNetFormer all reached more than 99%, and SeaIceNet also performed well in the recognition of these four categories, reaching 99.6464%, 99.8859%, 99.5568%, and 99.8859%, respectively. In the recognition of ThinFI, although there are still three models with recognition accuracy exceeding 99%, BANet has been opened up by ABCNet, UNetFormer, and SeaIceNet in this type of recognition, and SeaIceNet still reaches 99.7037% in this type of recognition. In the identification process of MedFI and ThickFI, we found that the recognition of the eight models showed a significant decline, but the stability of our proposed model was at least 3.7% and 4.8% higher than the other models, reaching 94.4655% and 91.7393%, respectively.

V. CONCLUSION

This article proposes a global–local detail-guided transformer method for sea ice segmentation. In the proposed SeaIceNet, the DGAH module extracts the vertical and horizontal information of the image through the dual global-attention mechanism, which enhances the global information extraction ability of the image. The GLFF module combines the local detail features extracted by the CNN module and the global structure features extracted by the Transformer module to enhance the detail enhancement capability of the model. In addition, the designed DGD enables the feature decoding process to retain detailed information about the sea ice edge region and improve the accuracy of semantic segmentation. Experiments on the generated sea ice dataset show that the proposed SeaIceNet achieves improved performance in sea ice segmentation and recognition in optical remote sensing images. However, it is not possible to separate the sand snow from the ice, and relevant methods will be explored in the follow-up study.

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